DD–G–426a
21 JULY 1954
SUPERSEDING
Int. Fed. Spec. DD–G–00426 (GSA–FSS)
2 January 1953 and
Fed. Spec. DD–G–426
20 June 1941

FEDERAL SPECIFICATION

GLASSES, COVER, (FOR) MICROSCOPY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** — This specification covers microscope cover glasses for microscopical observation.

1.2 **Classification.**

1.2.1 *Types, sizes, and thicknesses.*—Cover glasses covered by this specification shall be of the following types, sizes, and thicknesses, as specified:

Type I.—For blood counting chambers
Size 1.—20 mm. by 26 mm.
Thickness A.—0 4 mm. thickness (light)
Thickness B.—0 6 mm. thickness (heavy)

Type II.—For microscopy
Size 1.—square, 18 mm by 18 mm.
Size 2 —square, 22 mm. by 22 mm.
Size 3 —square, 24 mm. by 24 mm.
Size 4.—square, 25 mm. by 25 mm.
Size 5.—rectangle, 22 mm by 30 mm.
Size 6 —rectangle, 22 mm by 40 mm.
Size 7 —rectangle, 22 mm. by 50 mm.
Size 8.—round, 18 mm. in diameter
Size 9.—round, 25 mm. in diameter
Thickness C.—0.085 mm. to 0.13 mm. in thickness (No. 0)
Thickness D.—0.13 mm. to 0.17 mm. in thickness (No. 1)
Thickness E.—0.17 mm. to 0.25 mm. in thickness (No. 2)

2. APPLICABLE SPECIFICATION

2.1 There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Service Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo , Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies)

2.2 The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

Department of Commerce — Bureau of Standards Scientific Paper No. 507:

A new Interference Apparatus for Testing Haemacytometers.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also widely distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.—Unless otherwise specified, material shall be colorless transparent glass. No pink color shall be visible through 24 superimposed cover glasses when tested as specified in 4.2.3.

3.2 Physical requirements.

3.2.1 *Type I, for blood counting chambers.*

3.2.1.1 *Planeness.* — The surface of cover glasses for blood counting chambers shall not deviate on both sides from a true plane by more than 0.002 mm., when tested as specified in 4.2.1. When specified in the invitation for bids, each cover glass shall bear the Bureau of Standards certification.

3.2.1.2 *Tolerances.*—A tolerance of plus or minus 1.0 mm. will be allowed on the size measurements and a plus or minus 0.06 mm. on the thickness measurements.

3.2.1.3 Cover glasses shall be packaged 50 per box.

3.2.2 *Type II, for microscopy.*

3.2.2.1 *Thickness.* — The following tolerances will be allowed on the thickness measurements:

Thickness C (No. 0) cover glasses shall be no less than 0.085 mm. and no greater than 0.13 mm. in thickness.

Thickness D (No. 1) cover glasses shall be no less than 0.13 mm. and no greater than 0.17 mm. in thickness.

Thickness E (No. 2) cover glasses shall be no less than 0.17 mm. and no greater than 0.25 mm. in thickness.

3.2.2.2 *Dimensions.*—A tolerance of plus or minus 1.0 mm. will be allowed on the size dimensions of type II cover glasses.

3.2.2.3 The number of cover glasses packaged per box shall be in accordance with table I.

TABLE I.

Size	Dimensions, millimeters	Number for each thickness		
		C (No. 0)	D (No. 1)	E (No. 2)
1	18 x 18	165 ± 8	115 ± 6	88 ± 6
2	22 x 22	109 ± 5	77 ± 5	59 ± 4
3	24 x 24	93 ± 5	65 ± 5	49 ± 4
4	25 x 25	86 ± 5	60 ± 5	45 ± 4
5	22 x 30	81 ± 5	57 ± 5	43 ± 4
6	22 x 40	61 ± 5	43 ± 4	32 ± 3
7	22 x 50	49 ± 4	34 ± 3	26 ± 3
8	18 (diameter)	210 ± 10	147 ± 7	112 ± 7
9	25 (diameter)	109 ± 5	76 ± 5	58 ± 4

Note.—Number of cover glasses per box has been chosen so that each box will weigh approximately ½ ounce.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

3.3 Workmanship.—Cover glasses shall be uniform as to size and thickness and shall be free from such visible defects as bubbles, seeds, blisters, striae, or optical imperfections which would impair their efficiency or adversely affect their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Not less than 10 percent of type I and 1 percent of type II cover glasses delivered shall be taken at random for test purposes.

4.2 Inspection and tests.

4.2.1 *Planeness.*—The planeness of type I cover glasses shall be tested with an interferometer as outlined in Bureau of Standards Scientific Papers, No. 507.

4.2.2 *Thickness.* — Both types shall be measured with a standard cover glass gage.

4.2.3 *Solubility test.*—Twenty-four cover glasses shall be cleaned by immersion for 1 minute in distilled water, removed and air-dried for 1 minute in a vertical position. This cleaning operation shall be performed three times, using a new quantity of distilled water for each immersion. Half fill an alkali-resistant 600-ml Erlenmeyer flask with distilled water and boil for 10 minutes. Allow water to cool for 3 minutes, and then decant until the volume in the flask is approximately 100 ml. Keeping the flask covered, let stand until the water has cooled to approximately 70° C. Add 0.2 ml. of U S P. standard phenolphthalein solution and continue to cool to 60° C. Add the cover glasses, one at a time. The cover glasses shall be superimposed by tilting and gently swirling the flask. Let solution cool toward room temperature without agitation for 1 hour. Absorption of CO_2 from the atmosphere must be avoided during the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Type I.*—Unless otherwise specified, each cover glass shall be wrapped in soft paper and packaged in a suitably sized box. After the package is marked (see 5.3), it shall be wrapped and heat-sealed in a transparent moisture-resistant material. Packages shall contain but one thickness of cover glasses.

5.1.2 *Type II.*—Unless otherwise specified, the quantities of cover glasses specified in 3.2.2.3, weighing approximately ½ ounce, shall be wrapped and heat-sealed in a moisture-resistant material and packaged in a suitably sized box. After the package is marked (see 5.3), it shall be wrapped in a moisture-resistant material. Packages shall contain only one thickness and one size of cover glasses.

5.2 Packing.—Unless otherwise specified, cover glasses (packages or cartons) shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carrier, for safe transportation at the lowest rate, to the point of delivery.

5.3 Marking.

5.3.1 *Packages and cartons.*—Unless otherwise specified, each package (5.1.1 and 5.1.2) and cartons, if used, shall be marked to show the name of the item, the quantity contained therein, the thickness and size, and the name of registered trade-mark of the manufacturer or contractor.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the item, the quantity contained therein as defined by the contract or order under which shipment is made, the number of the contract or order, the type, size, and thickness, and the gross weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Purchasers should specify type, thickness, and size of cover glasses required and should exercise any desired options offered herein. (See 1.2.1, 3.1, 5.1, 5.2, 5.3, and 7.2.5.)

6.2 This specification covers the following items appearing in the Armed Services Catalog of Medical Materiel:

4–189–000 Cover Glass, Micro, 22 mm. Square ½ oz.
4–190–000 Cover Glass, Micro, 22 by 40 mm., ½ oz.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 Transportation description.

6.5.1 *The proper transportation description, carload and truckload weights to be added are as follows:*

Glass, Microscopical Cover
Carload weight 36,000 pounds
Truckload weight 36,000 pounds.

Patent notice.—The furnishing by the Government of specifications, including drawings or other data in connection therewith, does not in any way grant the recipient of such material, or any other person, a license or right to manufacture, use, or sell any patented or unpatented invention that may in any way be related thereto

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–621—Boxes, Wood, Nailed and Lock-Corner.
LLL–B–631—Boxes; Fiber Corrugated (for Domestic Shipment).

Military Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
JAN–P–133—Packaging and Packing for Overseas Shipment — Boxes, Set-Up, Paperboard.
MIL–L–10547—Liners, Case, Waterproof.

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

7.2.2 *Packaging for domestic or overseas shipment.*

7.2.2.1 *Unit packaging.*—The number of cover glasses specified in 3.2.2.3 constitutes one unit. Each unit shall be packed in a carton of appropriate size constructed in accordance with the requirements of Military

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.

Specification JAN–P–133. Carton shall be adequately secured to prevent accidental opening.

7.2.2.2 *Intermediate packaging.*—Twenty unit cartons (20 units) of one stock number shall be packaged in a carton of appropriate size constructed in accordance with the requirements of Military Specification JAN–P–133 or Federal Specification LLL–B–631. Each carton shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.* — Fifty intermediate cartons (1,000 units) of one stock number shall be overpacked in an exterior container designed for type 3 load. Construction, closure, and strapping (when required) shall be in accordance with the requirements of Federal Specification NN–B–621.

7.2.3.2 *For overseas shipment.* — Fifty intermediate cartons (1,000 units) of one stock number shall be overpacked in an exterior container designed for type 3 load. Construction, closure, and strapping shall be in accordance with the requirements of Military Specification JAN–P–106.

7.2.3.2.1 *Case liner.* — Each box shall be lined with a waterproof case liner conforming to the requirements of Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

7.2.3.3 *Packing variation permitted.* — If the number of units in any shipment is less than the specified number of units to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design which will insure safe delivery to destination.

7.2.4 *Marking.*

7.2.4.1 *Interior packages.* — Each unit carton and each intermediate carton shall be marked with stock number, nomenclature, unit, quantity, office identifying number, and name of contractor.

7.2.4.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5 *Ordering data.* — Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.3.1 and 7.2.3.2.)

MILITARY INTERESTS:

Army—M C
Navy—Md S
Air Force.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:42Z
Check the source to verify that this is the current version before use.